Evidence of particle acceleration in Astrophysical objects

(Lecture 6)

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Outline of this lecture

- Evidence of particle acceleration in Galactic objects
  - Supernova remnants
- Evidence in Extra-galactic objects
  - Active Galactic Nuclei
  - Gamma-ray bursts

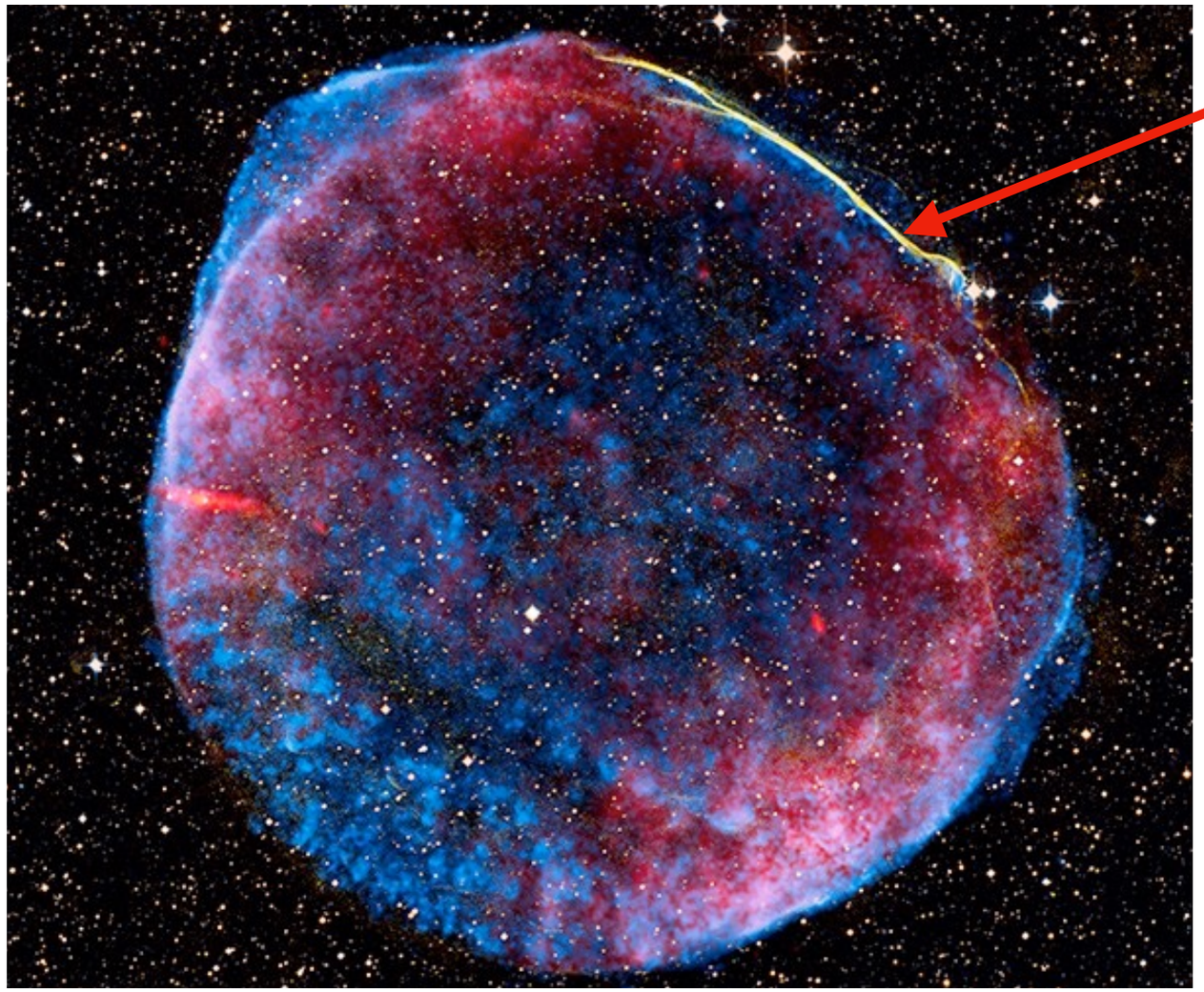
Galactic objects

Supernova remnants (SNRs): Remains of supernova explosions

- SNRs are considered as the most potential sources of high-energy cosmic rays in the Galaxy.
- Observed across all frequencies from radio to X-rays to very-high-energy gamma rays.
- About 300 Galactic supernova remnants are known. Most of them have been discovered over the last decades with radio telescopes.
- A handful has been detected in X-rays.
- Only about 12 are known in very-high energy gamma rays.

Galactic objects

SN1006 supernova remnant



Distance: 2.2 kpc
Age: ~ 1000 yr

Chandra telescope image

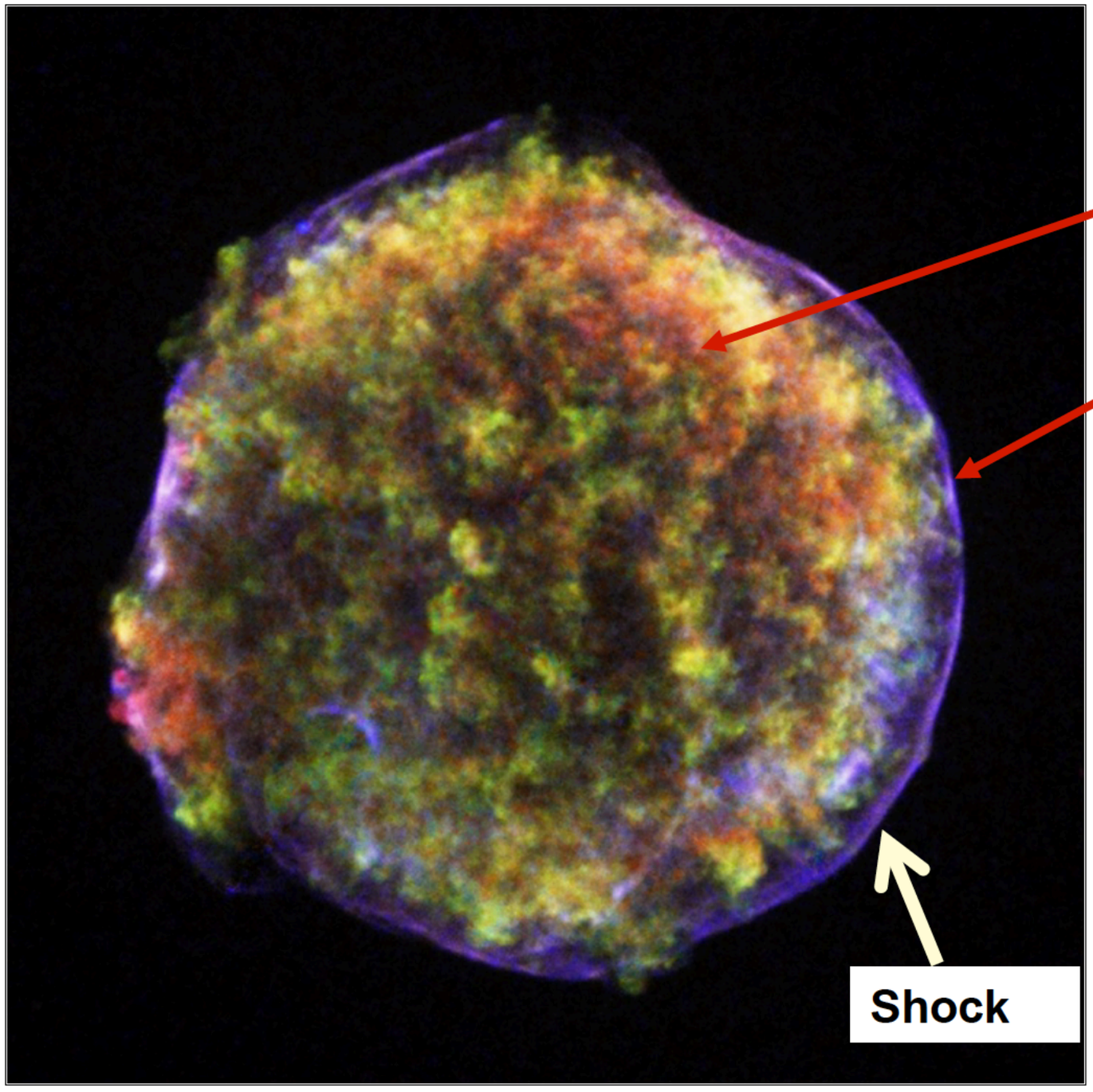
*Non-thermal X-rays from high-energy electrons: Blue. The rim indicates the shock waves accelerating electrons.

*Thermal X-rays from multi-million degree hot gas: red/green.

*It was likely a white dwarf star that had been pulling matter off an orbiting companion star. When the white dwarf mass exceeded the stability limit (known as the Chandrasekhar limit), it exploded. Material ejected in the supernova produced tremendous shock waves that heated gas to millions of degrees and accelerated electrons to extremely high energies.

https://chandra.harvard.edu/photo/2005/sn1006/

Tycho's Supernova Remnant



http://chandra.harvard.edu/photo/2005/tycho/

Exploded in 1572 and studied by Tycho Brahe (3kpc)

This is a Chandra X-ray image

Shock heated gas inside 3000 km/s blast wave (filamentary blue)

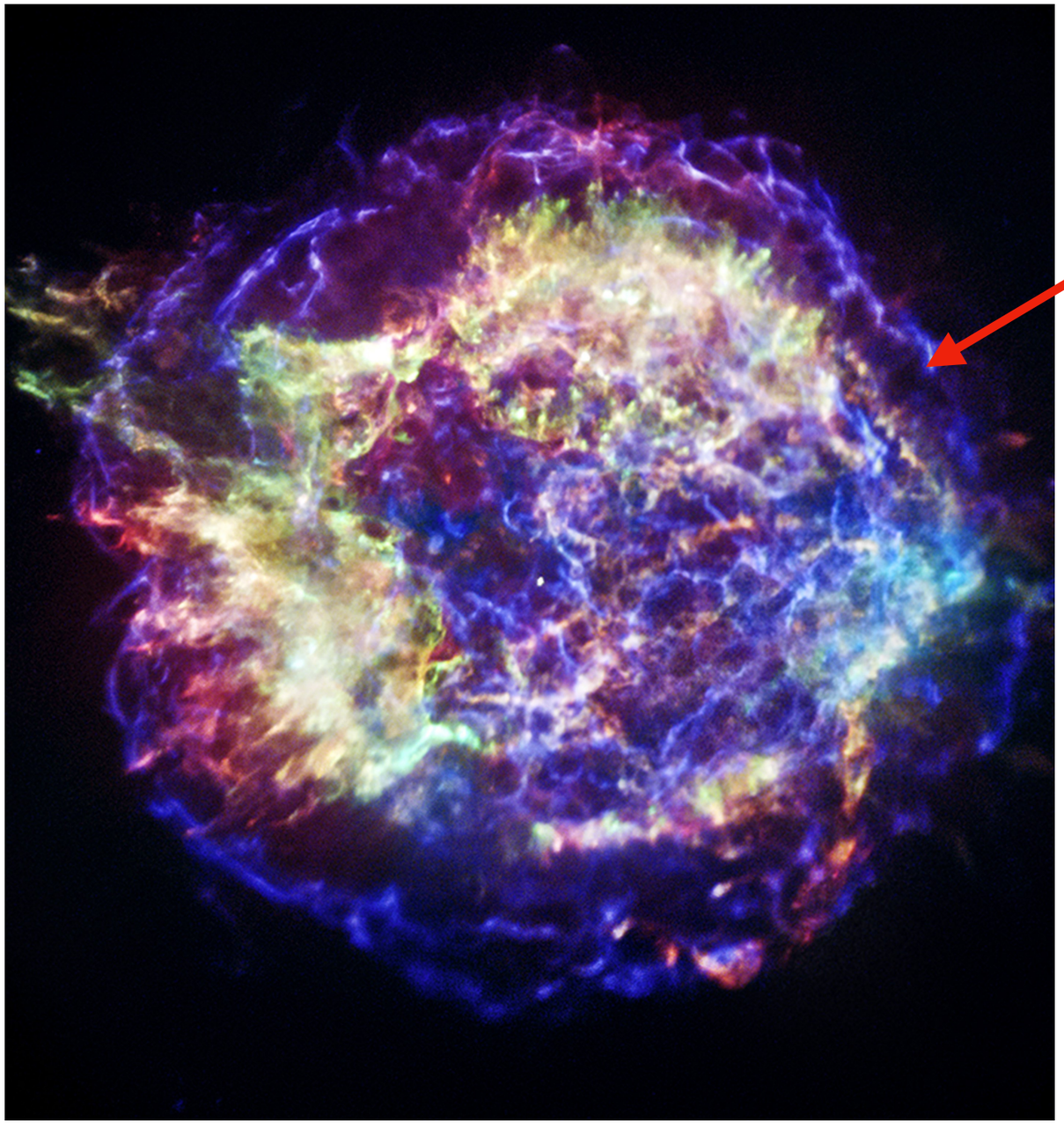
Blue is nonthermal X-ray emission (synchrotron) from shock accelerated relativistic electrons.

No doubt that TeV electrons are produced by this shock !!

Evidence for TeV ions is less direct but very strong.

Galactic objects

Cassiopeia A supernova remnant



Distance: 3.4 kpc
Age: 400-500 yr

Chandra telescope image

*Non-thermal X-rays from high-energy electrons: Blue rim around the remnant. Shock/blast waves from the explosion produce high-energy electrons as it travels outward.

*Thermal X-rays from million degree heated gas: Silicon (red), sulfur (yellow), calcium (green), and iron (light purple).

*Radio emission Very Large Array: Dark purple, blue, and white

*Visible light from Hubble telescope: Orange colour.

https://chandra.harvard.edu/photo/2022/archives/

Galactic objects

Supernova remnant G327.1-1.1



Credits: NASA/CXC/GSFC/T. Temim et al.

Distance: ~ 9 kpc
Age: ~18,000 yr

Chandra telescope image

*Non-thermal X-rays: Faint red. Outward-moving, or forward, shock wave.

*Pulsar wind nebula: Blue.

*The pulsar wind nebula appears to have been distorted by the combined action of the reverse shock wave, which may have flattened it, and by the motion of the pulsar, which created a comet, or lobster-like tail. An asymmetric supernova explosion may have given a recoil kick to the pulsar, causing it to move rapidly and drag the pulsar wind nebula along with it.

https://www.nasa.gov/mission_pages/chandra/news

Galactic objects

Supernova remnant MSH 11-62



Credits: NASA/CXC/SAO/P. Slane et al.

Distance: ~ 6 kpc
Age: ~1300 yr

Chandra telescope image

*Non-thermal X-rays: Blue. High energy X-rays.

*Thermal X-rays: Red. Irregular shell of hot gas.

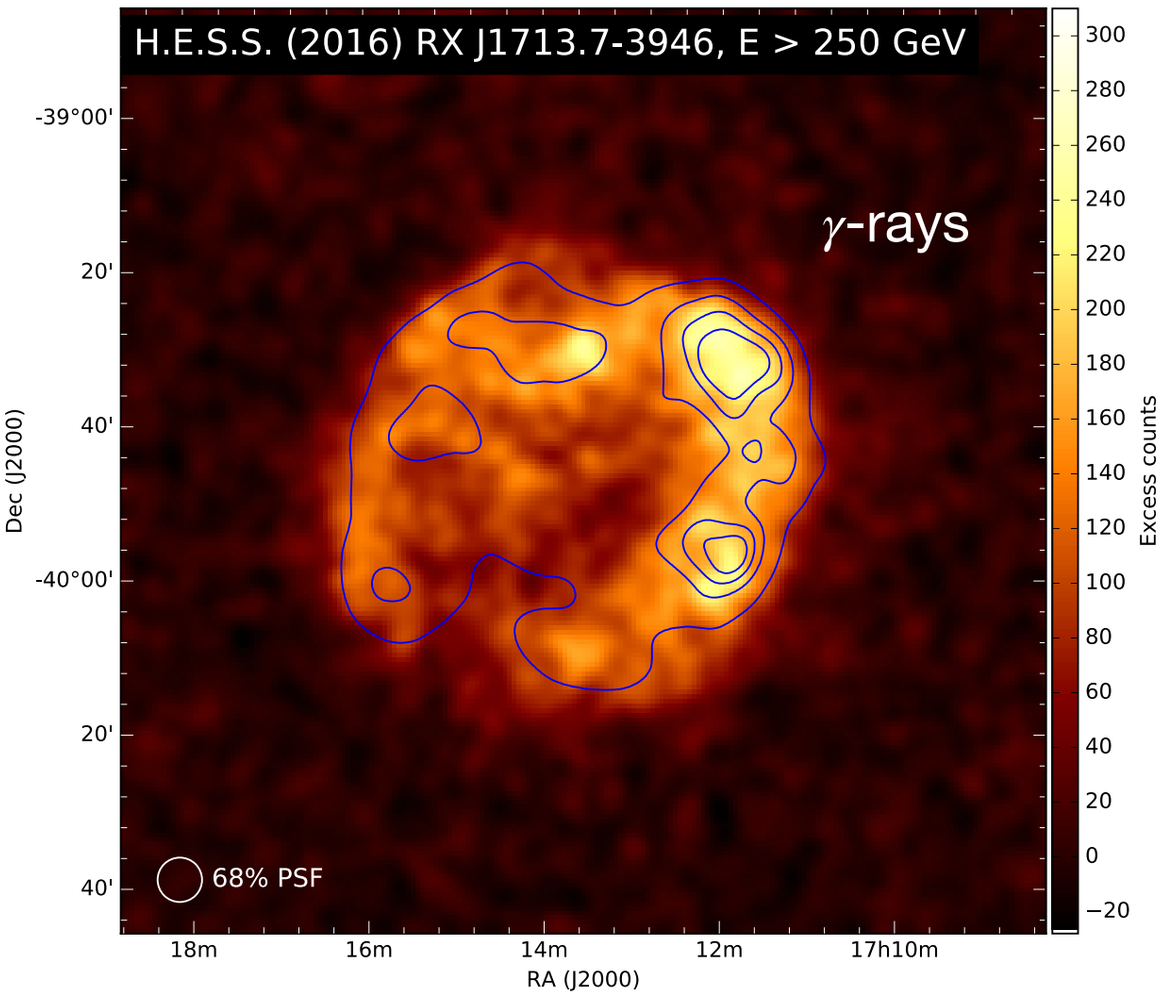
*Even though scientists have yet to detect any pulsations from the central object within MSH 11-62, the structure around it has many of the same characteristics as other pulsar wind nebulas. The reverse shock within MSH 11-62 appear to have begun to crush the pulsar wind nebula, possibly contributing to its elongated shape.

https://www.nasa.gov/mission_pages/chandra/news

Galactic objects

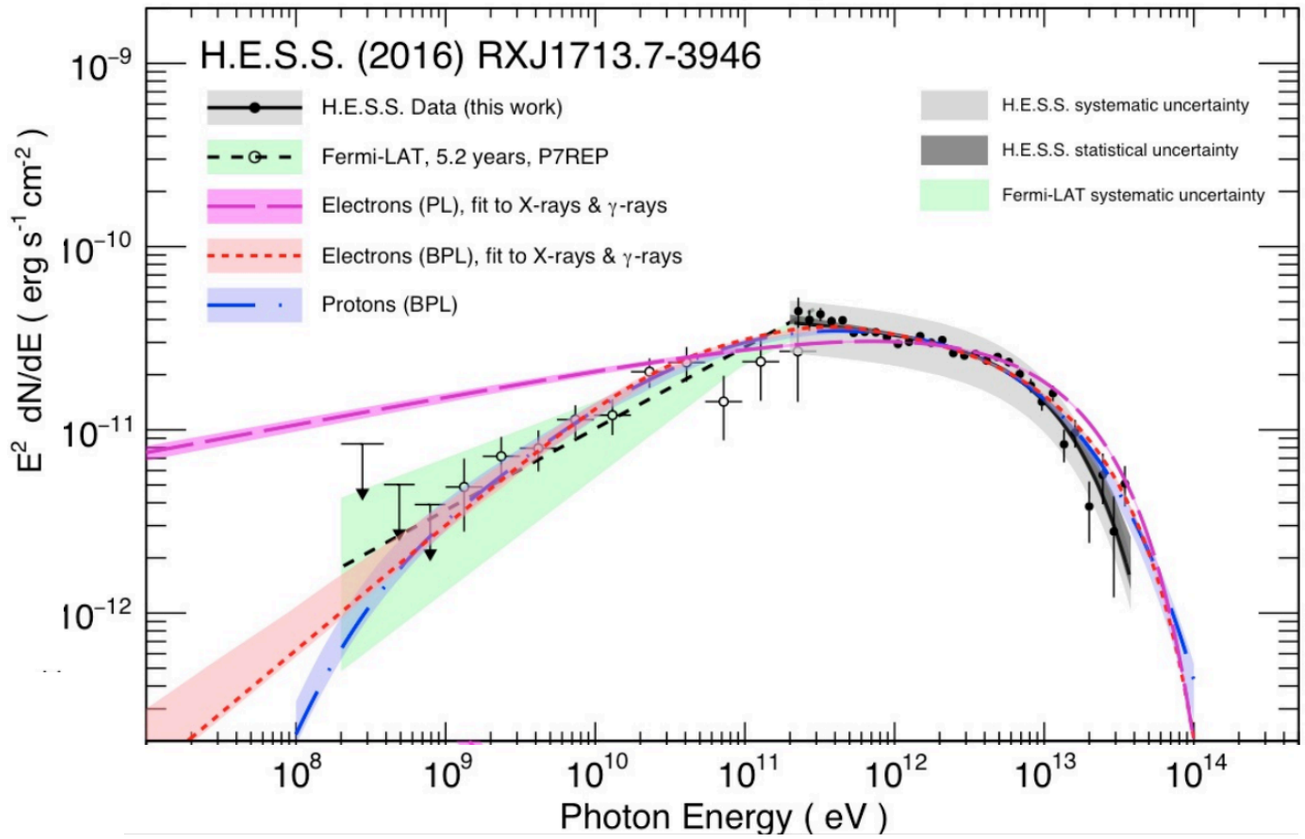
Supernova remnant RX J1713.7-3946

Distance: ~ 1 kpc
Age: ~1000 yr

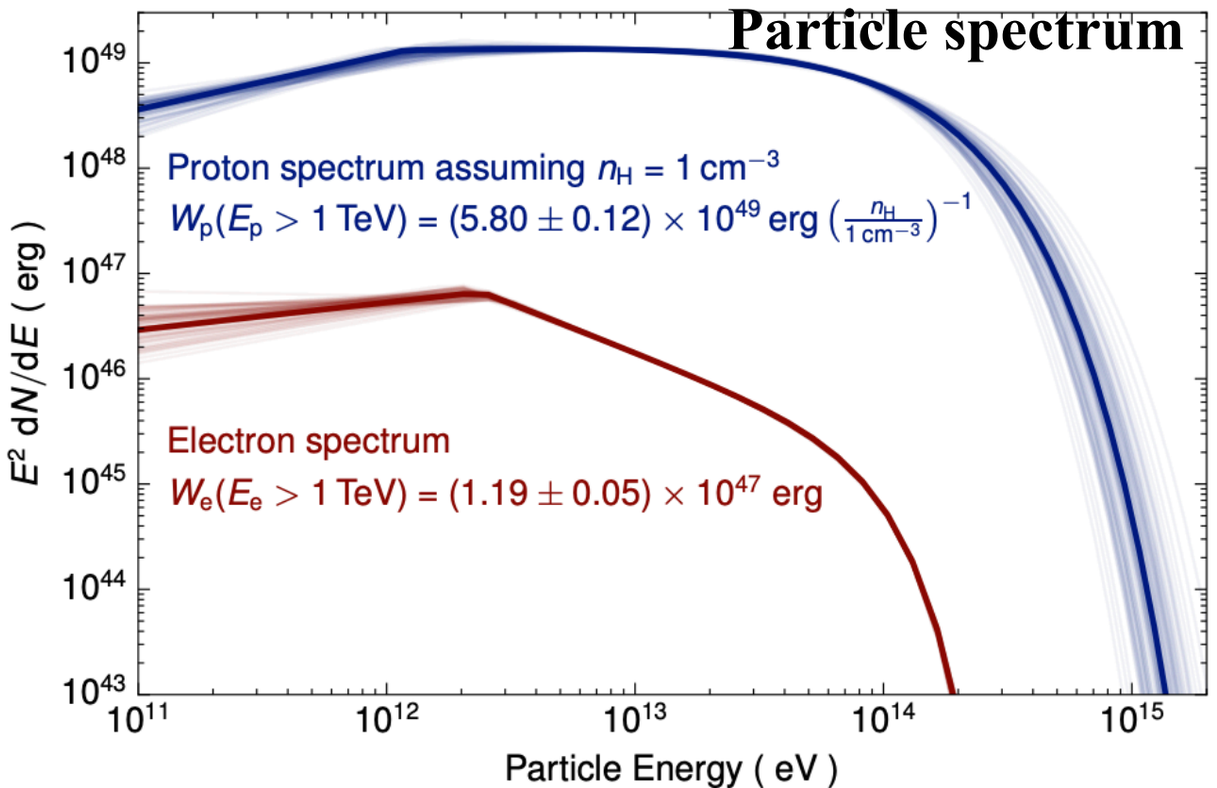


H.E.S.S Collaboration

Gamma-ray spectrum



Particle spectrum

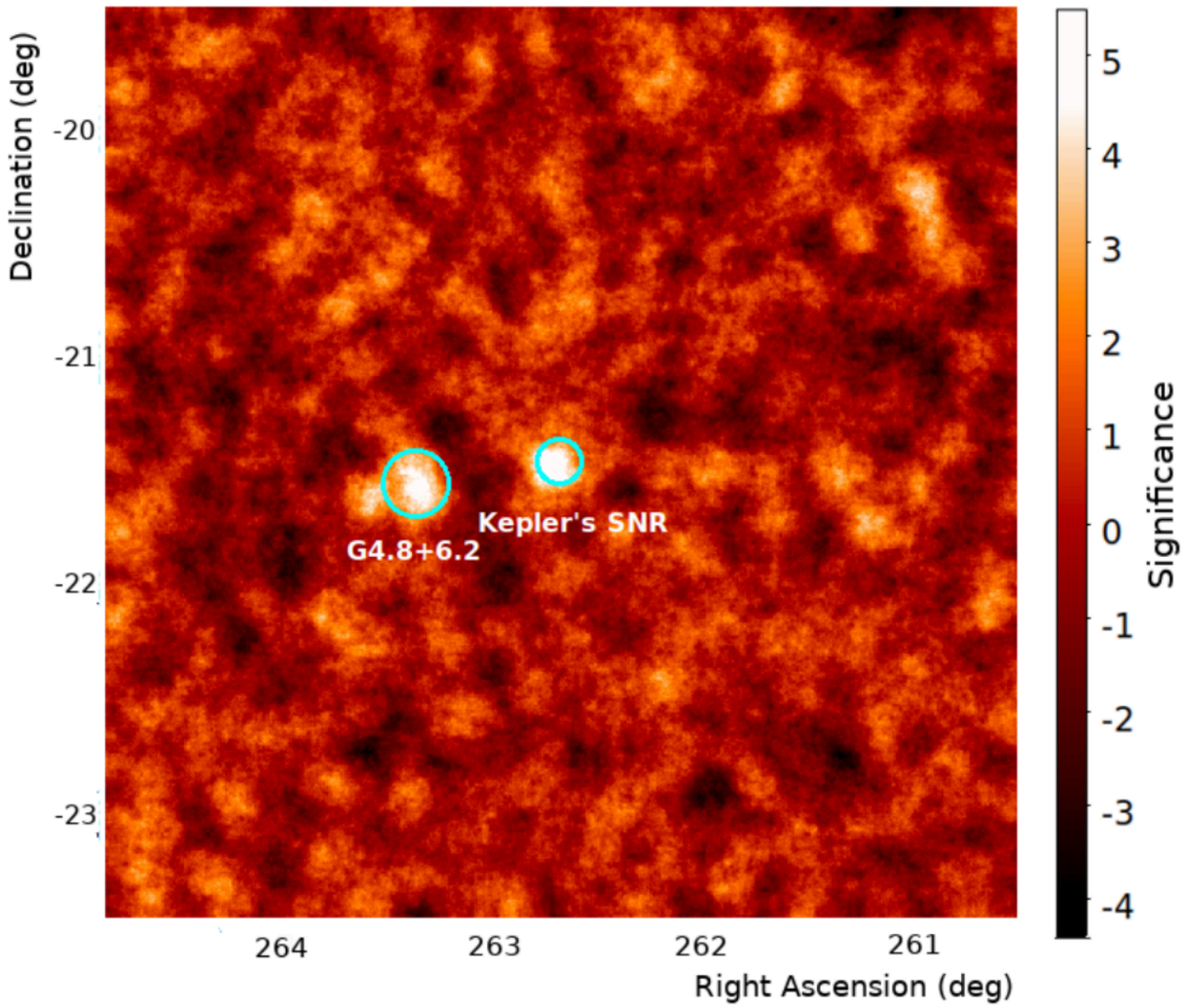


arXiv:1609.08671

Galactic objects

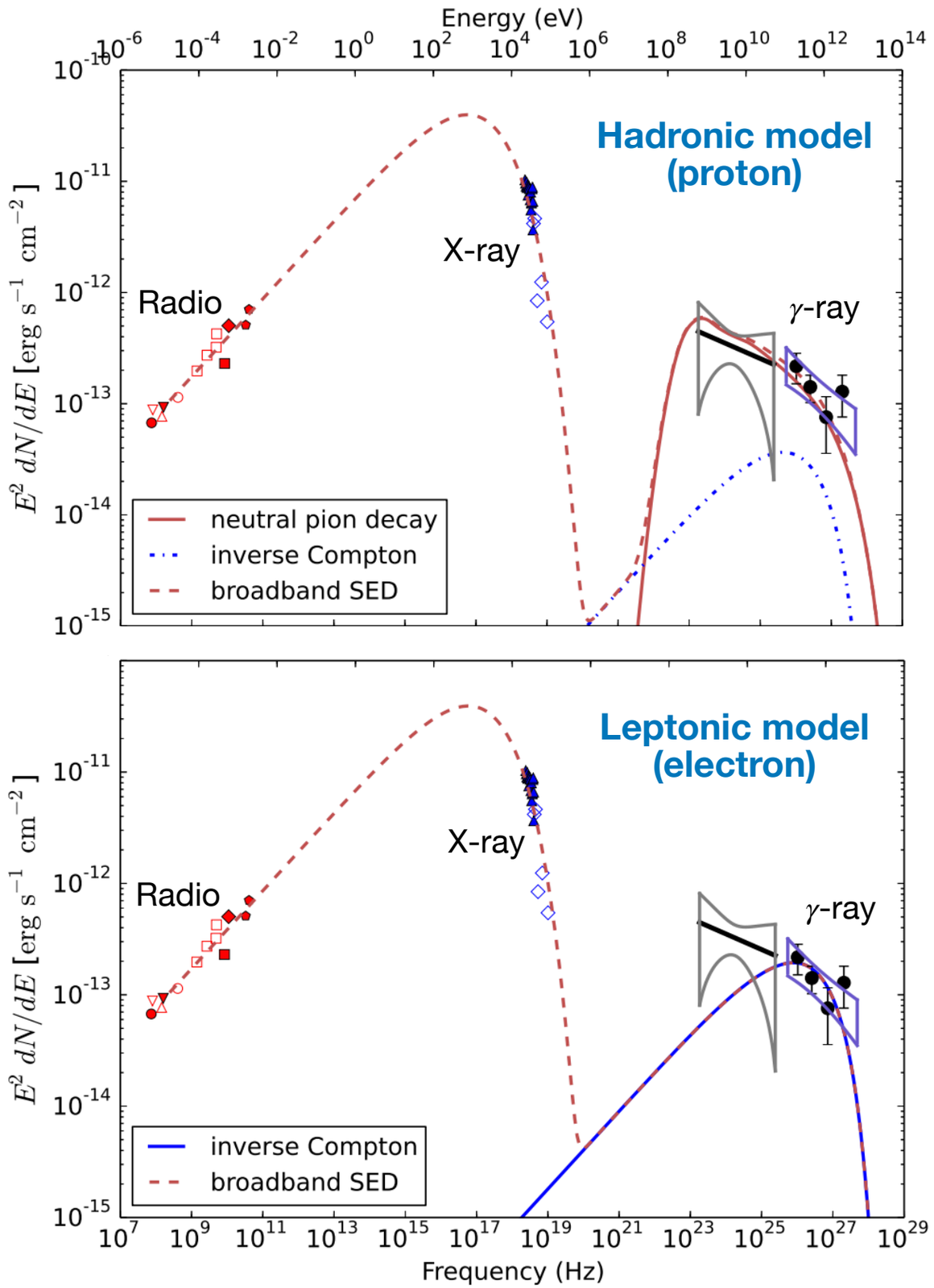
Kepler's Supernova remnant

Distance: ~ 6.1 kpc
Age: ~ 400 yr



H.E.S.S Collaboration

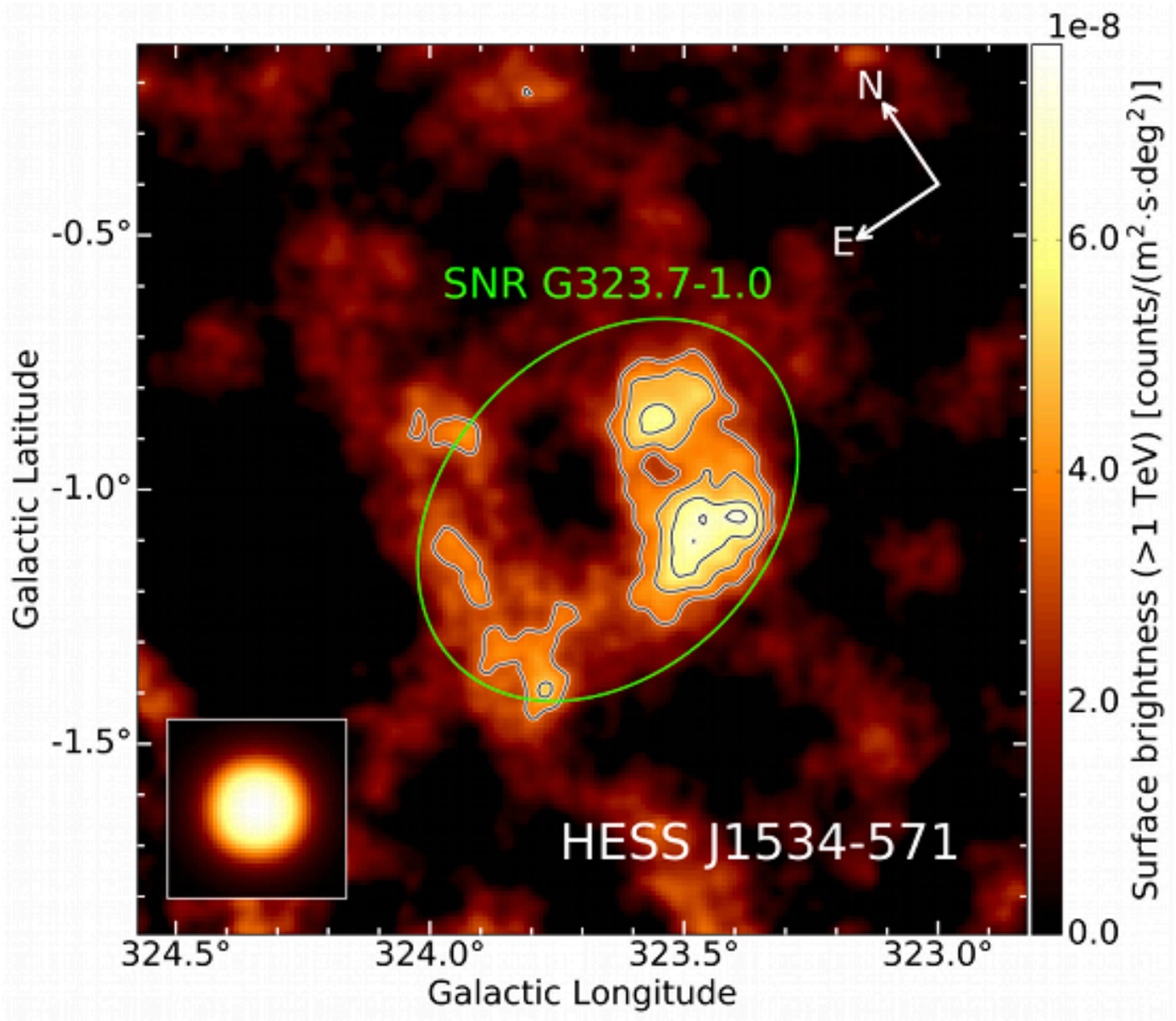
Multi-wavelength spectrum



https://arxiv.org/pdf/2201.05839.pdf

Galactic objects

HESS J1534-571 Supernova remnant



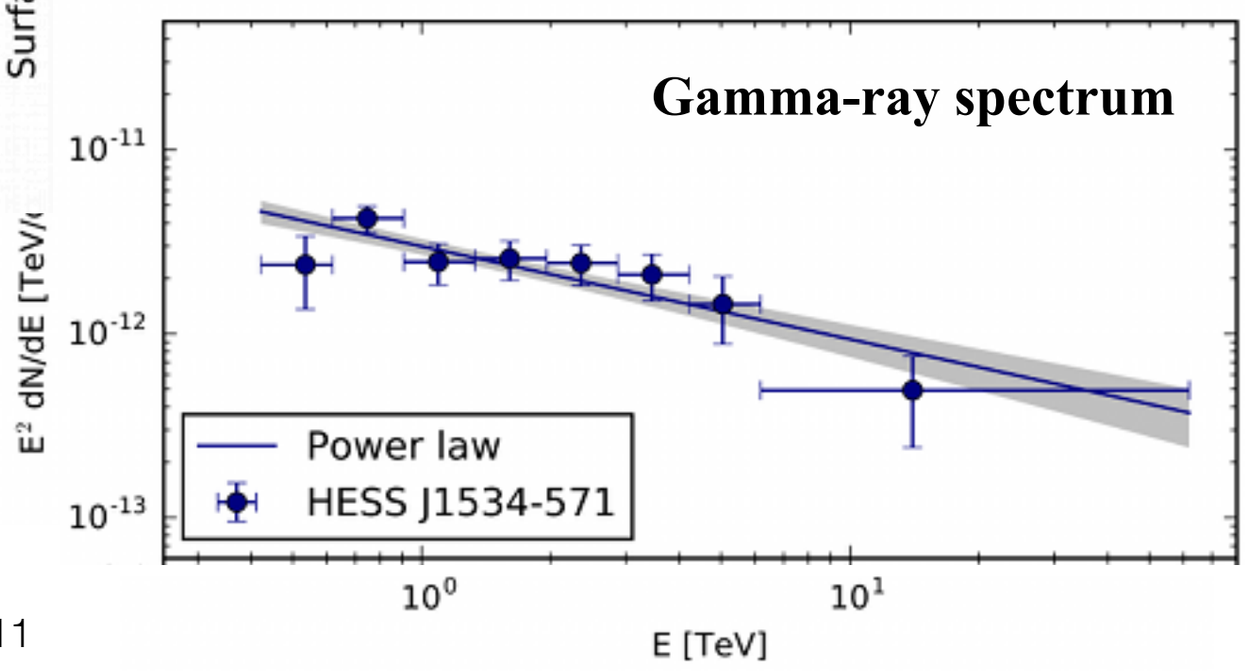
H.E.S.S Collaboration

* Very-high-energy gamma rays: Bright color/contour.

* Radio: Green ellipse. It denotes the extension of the radio SNR candidate counterpart G323.7-1.0.

*The inset on the bottom left denotes the point spread function of the TeV analysis.

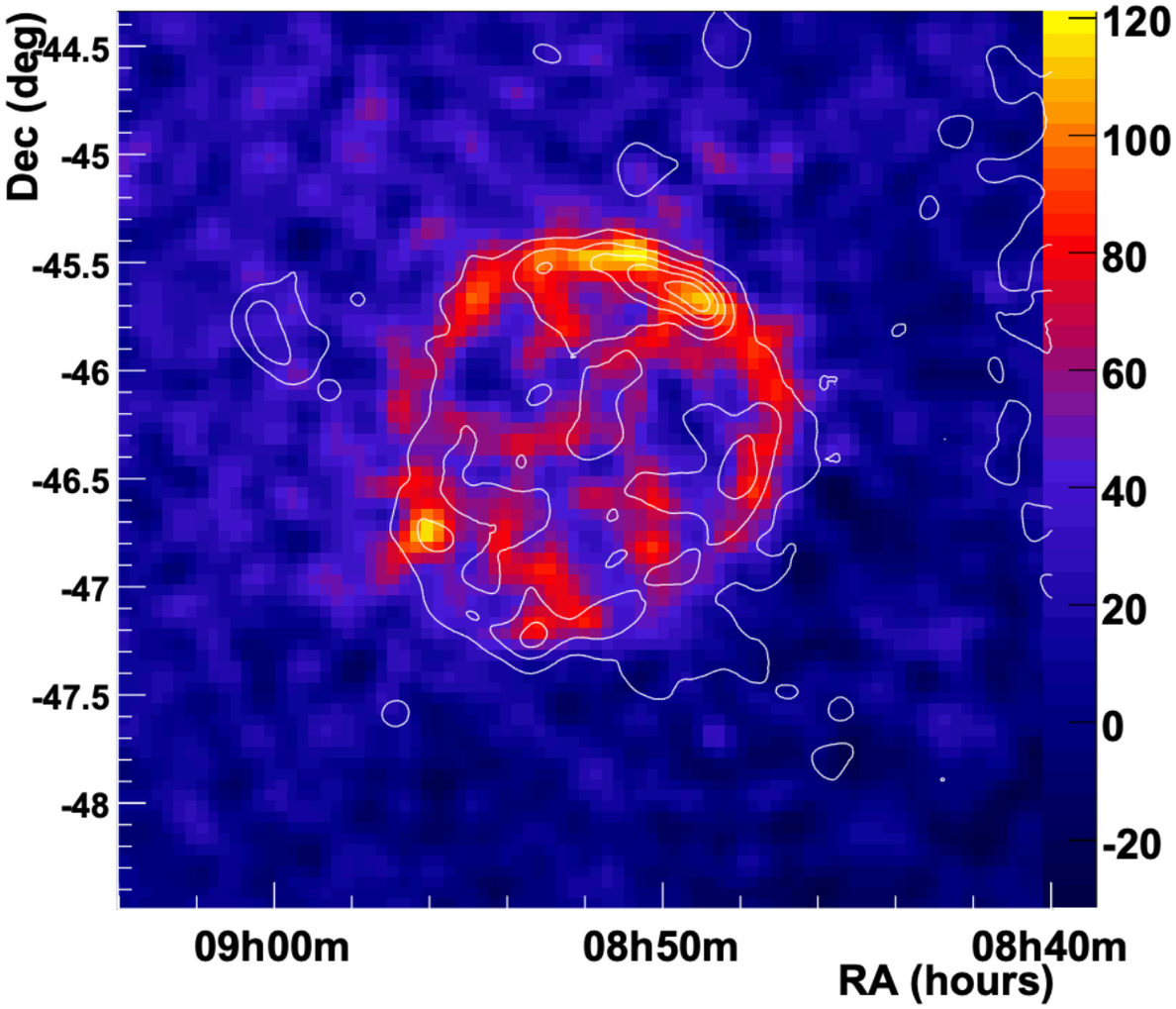
*A shell-type appearance is suggestive from the map.



https://www.mpi-hd.mpg.de/hfm/HESS/pages/home/som/2017/02/

Galactic objects

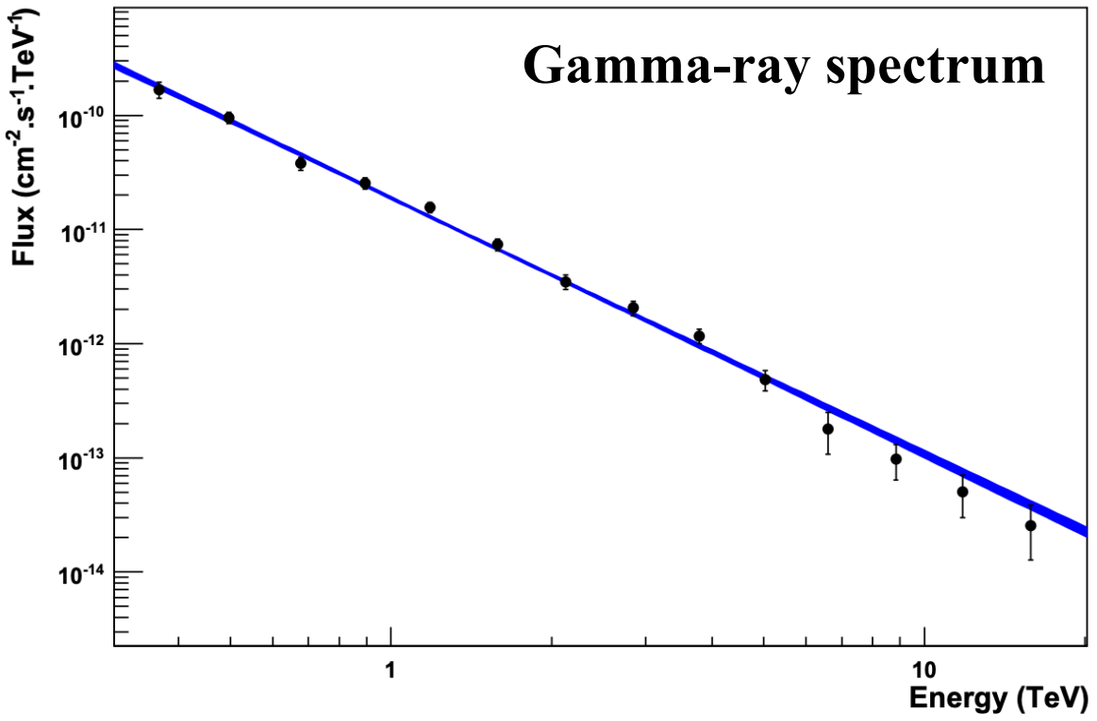
RX J0852.0-4622 Supernova remnant



H.E.S.S Collaboration

* Very-high-energy gamma rays: Bright color.

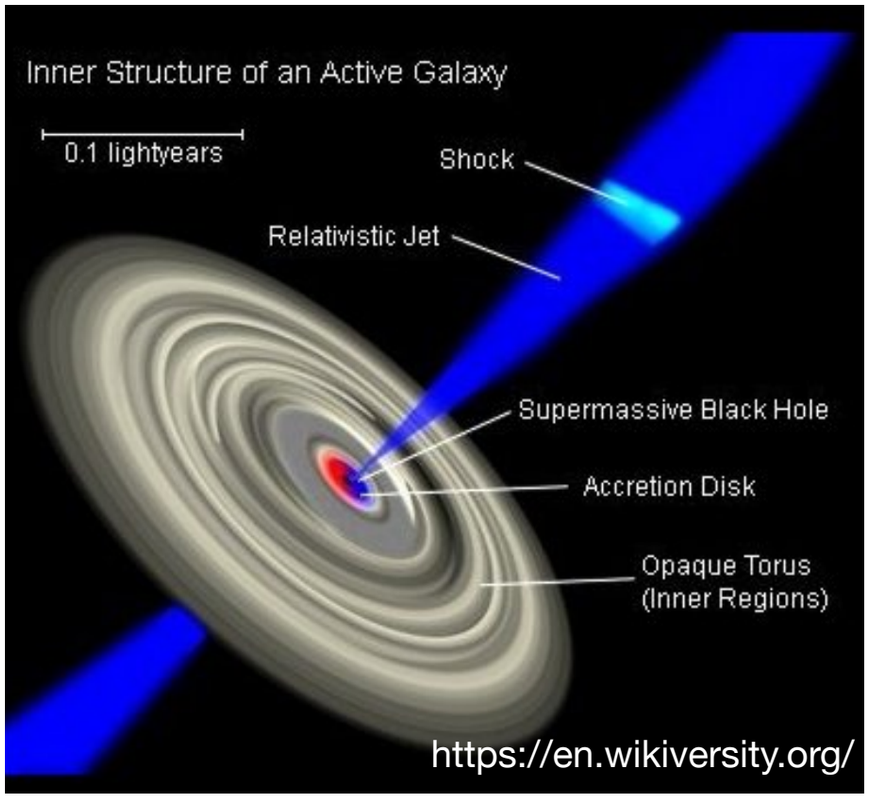
*X-rays from ROSAT All Sky Survey for energies higher than 1.3 keV: White contours.



https://arxiv.org/pdf/astro-ph/0612495.pdf

Extra-galactic objects

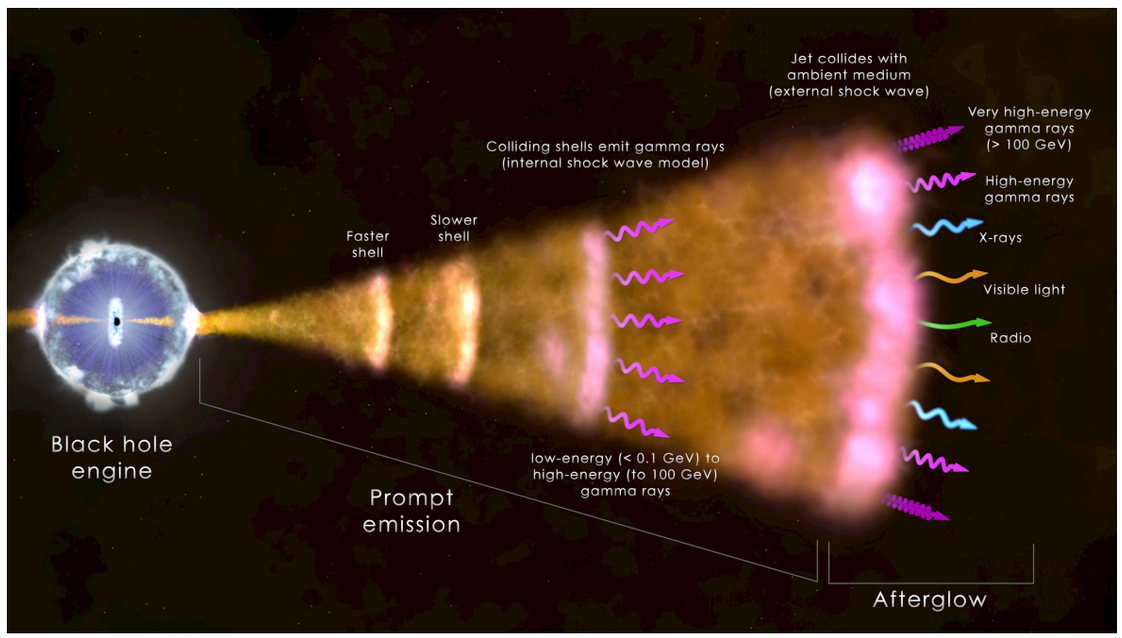
Active Galactic Nucleus (AGN)



*AGNs are the compact central region of galaxies which are 100 times more luminous and more massive than of our galaxy.
* Characterize by a pair of powerful jets, accretion disk and a supermassive black hole at the center.

Both are candidate sources of ultra high-energy cosmic rays.

Gamma-ray bursts (GRBs)



*GRBs are extremely energetic explosions that have been observed in distant galaxies.
*Most energetic and luminous events since the Big Bang.
*Bursts can last from ten milliseconds to several hours.
*After an initial short gamma-ray flash, a longer "afterglow" is usually emitted at longer wavelengths (radio to X-rays).

https://en.wikiversity.org/

Extra-galactic objects

*Extra-galactic objects are easily resolved at lower frequencies (radio, optical, X-rays).

*At very-high-energy gamma-rays, resolution is poor —> Almost all extra-galactic objects appear as point sources.

*As a result, most of the sources that appear spatially extended in very-high-energy gamma-ray images are Galactic sources, a few to tens of parsecs across and a few kilo-parsec away from Earth.

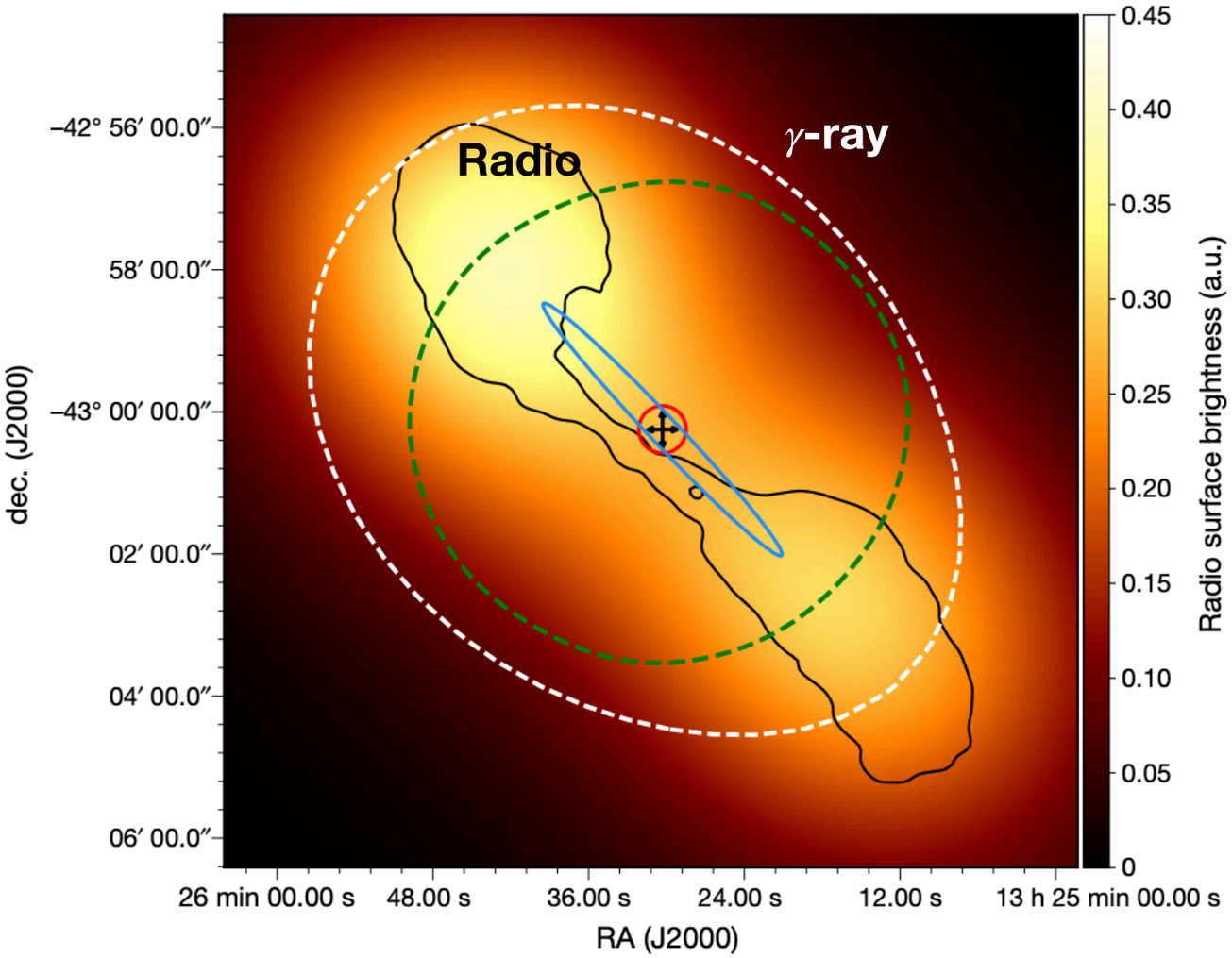
*Diameters of extragalactic objects have not been directly measured in the very-high-energy domain.

* A major breakthrough happens in 2020 with the HESS gamma-ray telescope observation of Centaurus A, an active galactic nuclei (The H.E.S.S. Collaboration: Nature, 2020, 582, 356–359).

* Centauras A is also special for cosmic-ray studies as it coincides with the position of one of the ultra-high energy cosmic rays detected by the Pierre-Auger Observatory.

Extra-galactic objects

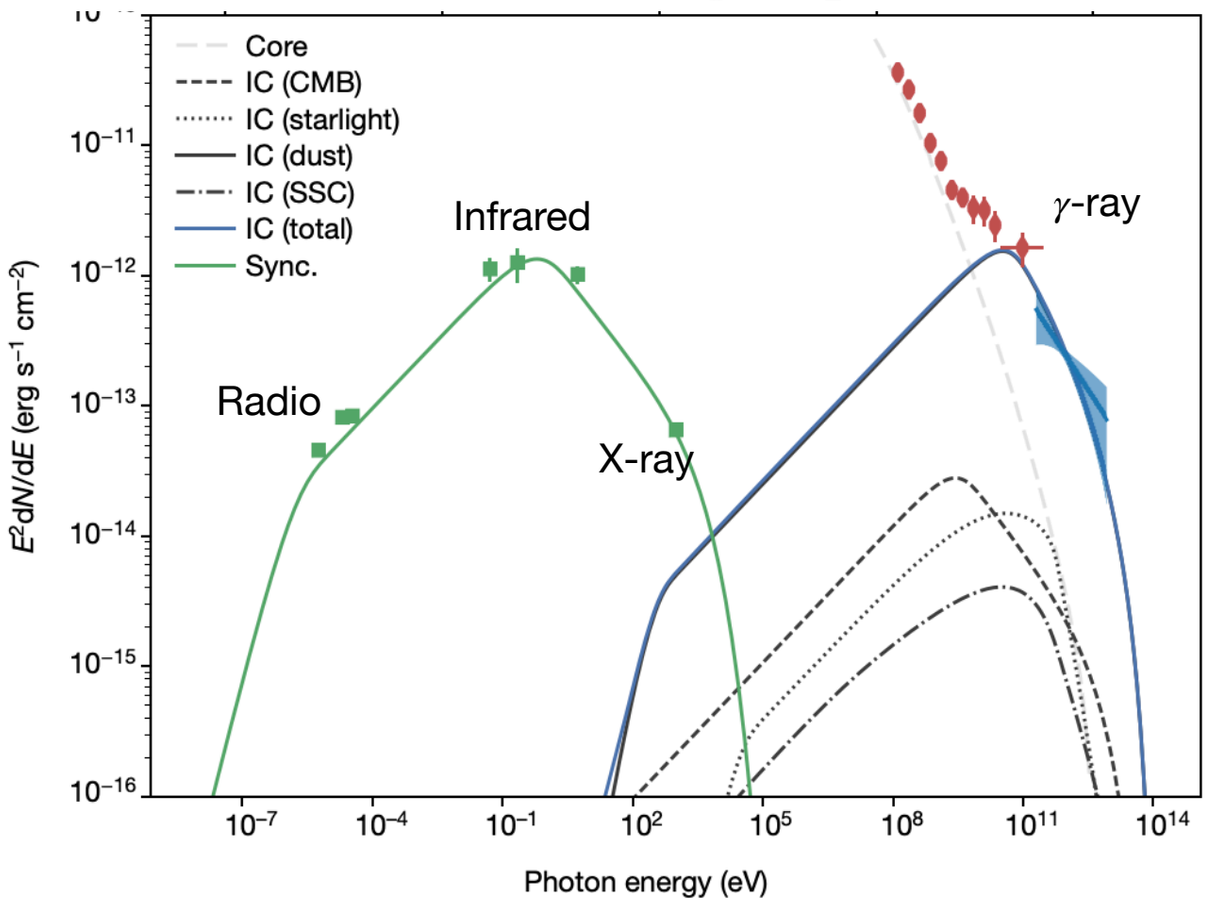
Centaurus A (Active Galactic Nuclei)



H.E.S.S Collaboration

* Closest known radio galaxy.
* Distance ~ 3.8 Mpc.
* Radio morphology exhibits kiloparsec-scale radio jet extending out from the nucleus.
* Very-high-energy TeV γ-ray emission spatially extended along the direction of the well-known radio-jet.

Multi-wavelength spectrum



The H.E.S.S. Collaboration: Nature, 582, 356–359 (2020)

Extra-galactic objects

Markarian 501 (Blazar)

Radio images from VLBA

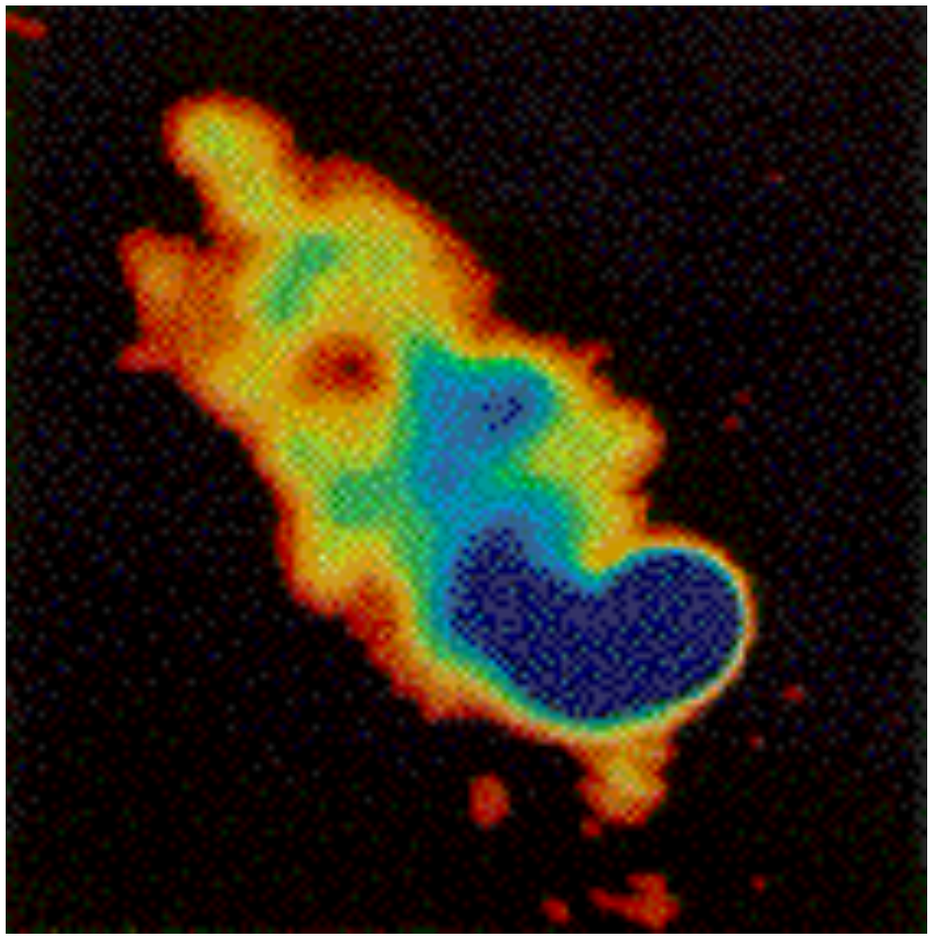


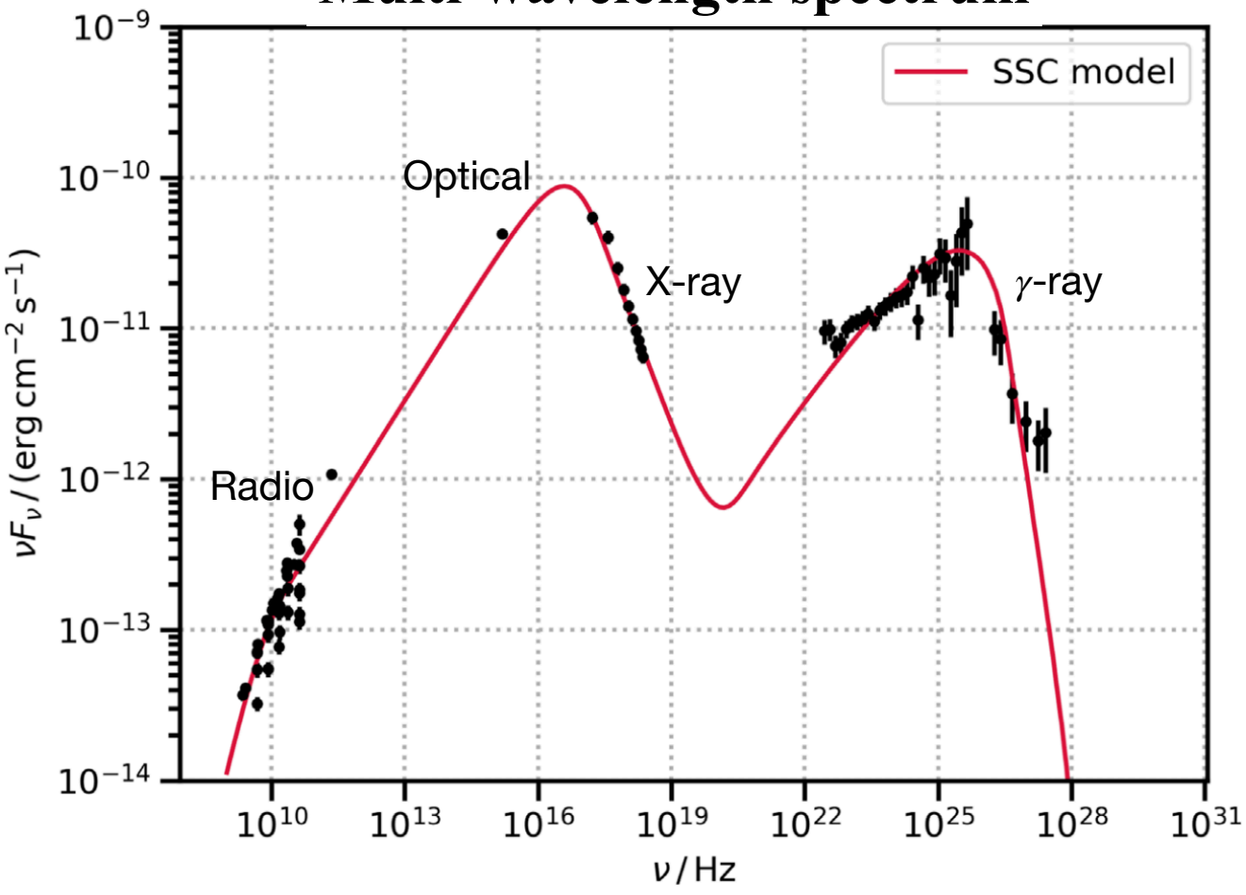
Image by J.M. Wrobel and J.E. Conway

Blazar: An active galactic nucleus with its relativistic jet pointing very close towards the Earth.

*Markarian 501 has an active nucleus, the source of energy for the jet of fast-moving particles seen in this VLBA image.

*VLBA images like this one help astronomers to understand the nature of jets, active galactic nuclei, and the black holes they may harbor.

Multi-wavelength spectrum



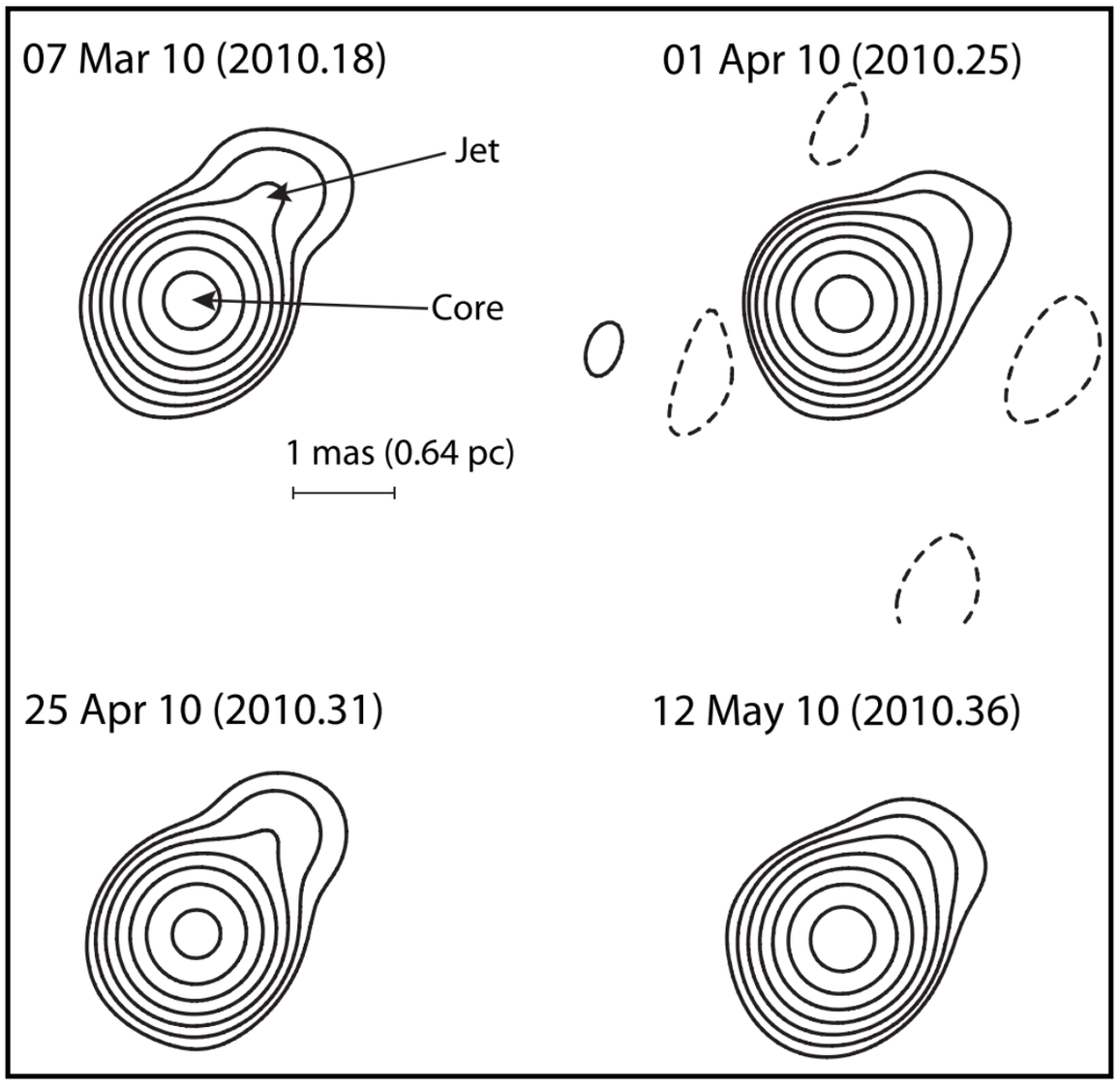
HAWC Collaboration, The Astrophysical Journal, 2022, 929, 125

http://www.vlba.nrao.edu/whatis/mark.html

Extra-galactic objects

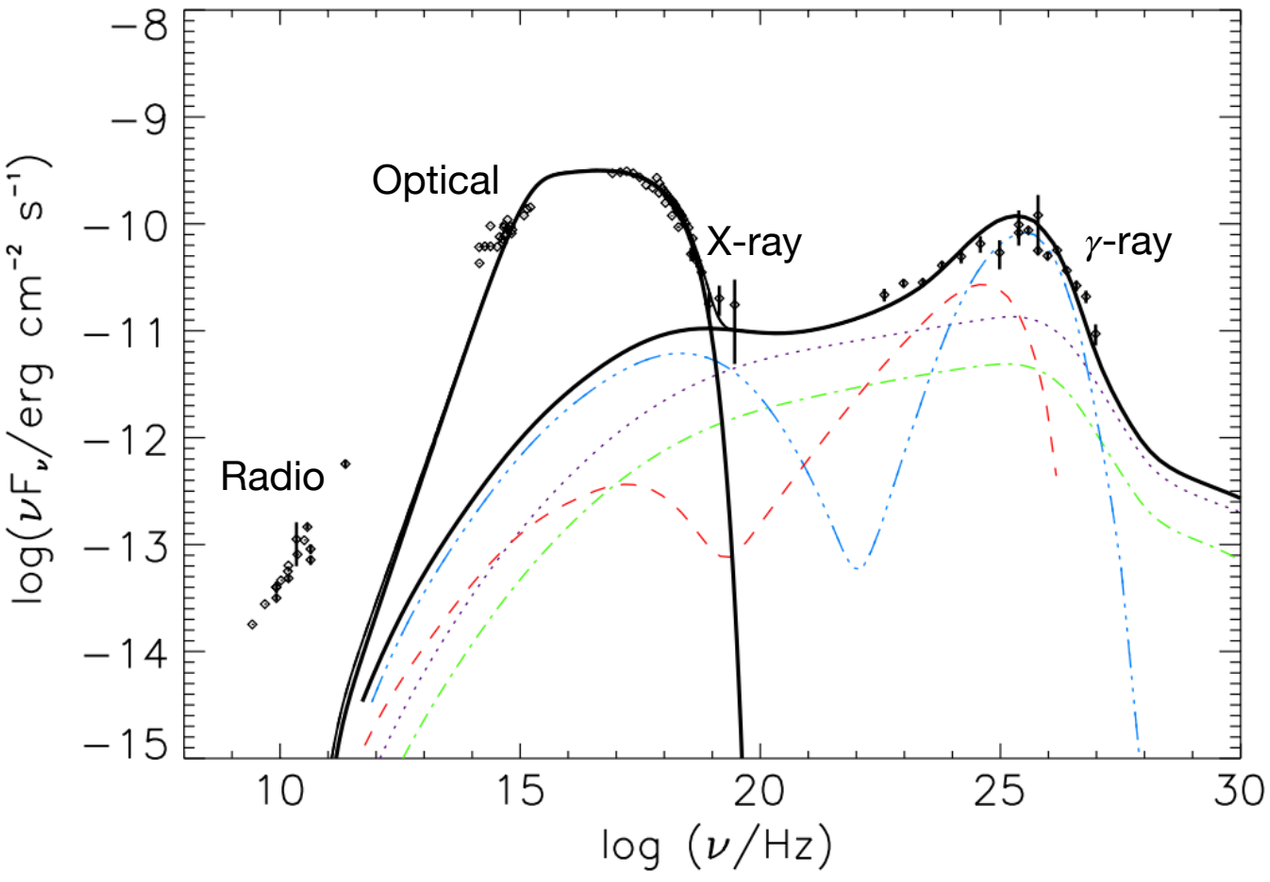
Markarian 421 (Blazar)

Radio images at different epoch



Niinuma et al., The Astrophysical Journal,2012, 759, 84

Multi-wavelength spectrum



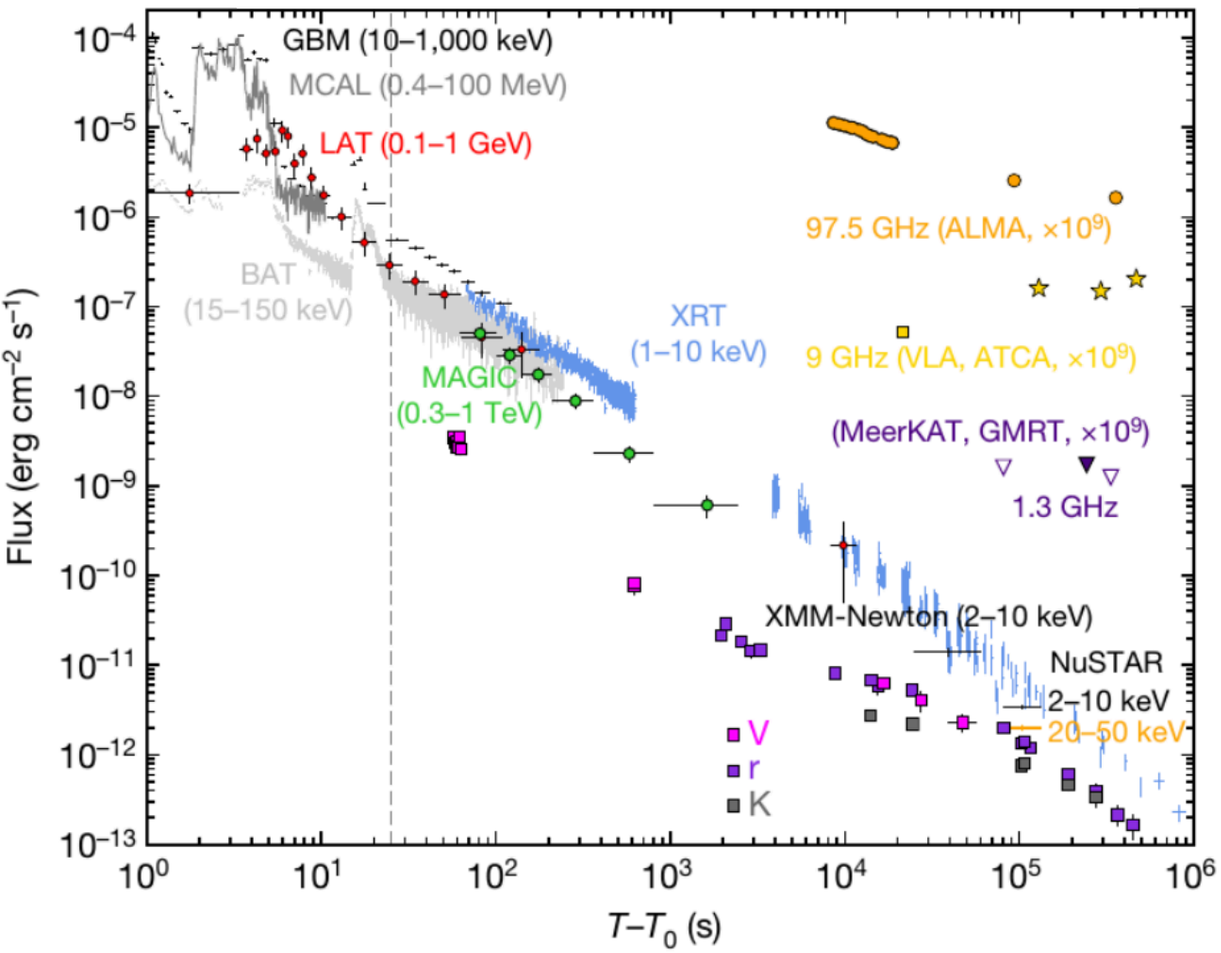
Fermi Collaboration, The Astrophysical Journal, 2011, 736, 131

Very Long Baseline Interferometry (VLBI)

Extra-galactic objects

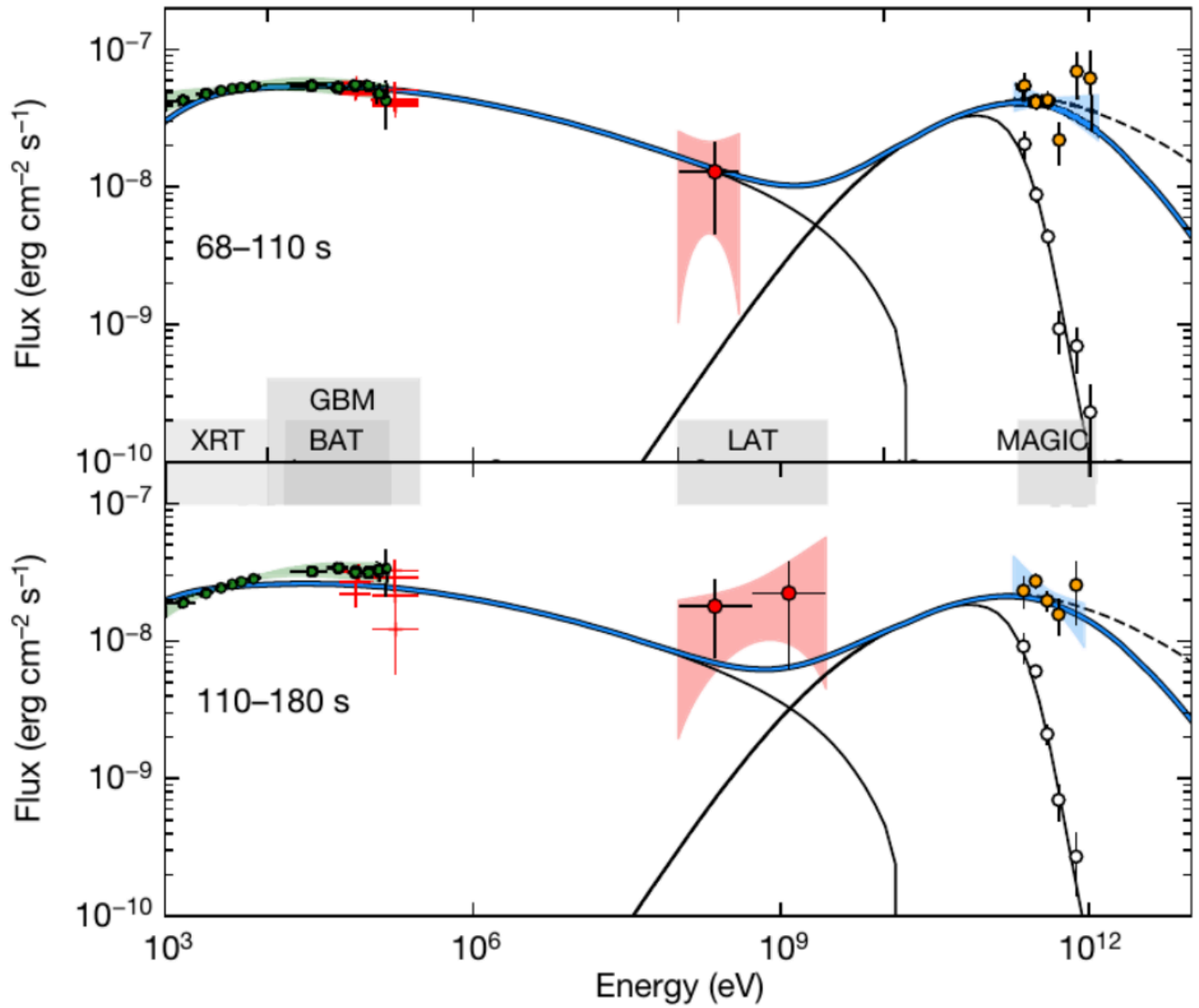
GRB 190114C

Light curve (Radio to TeV γ-rays)



First detection at TeV energies by the MAGIC telescope

Multi-wavelength spectrum

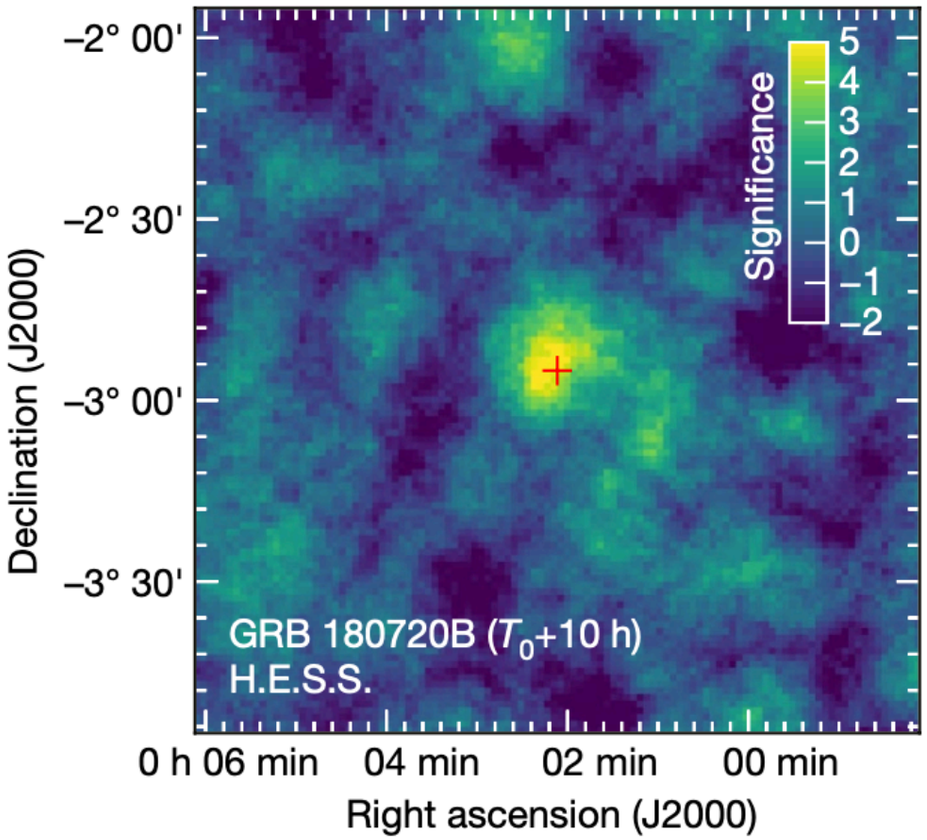


X-ray **LE γ-ray** **HE γ-ray** **VHE γ-ray**

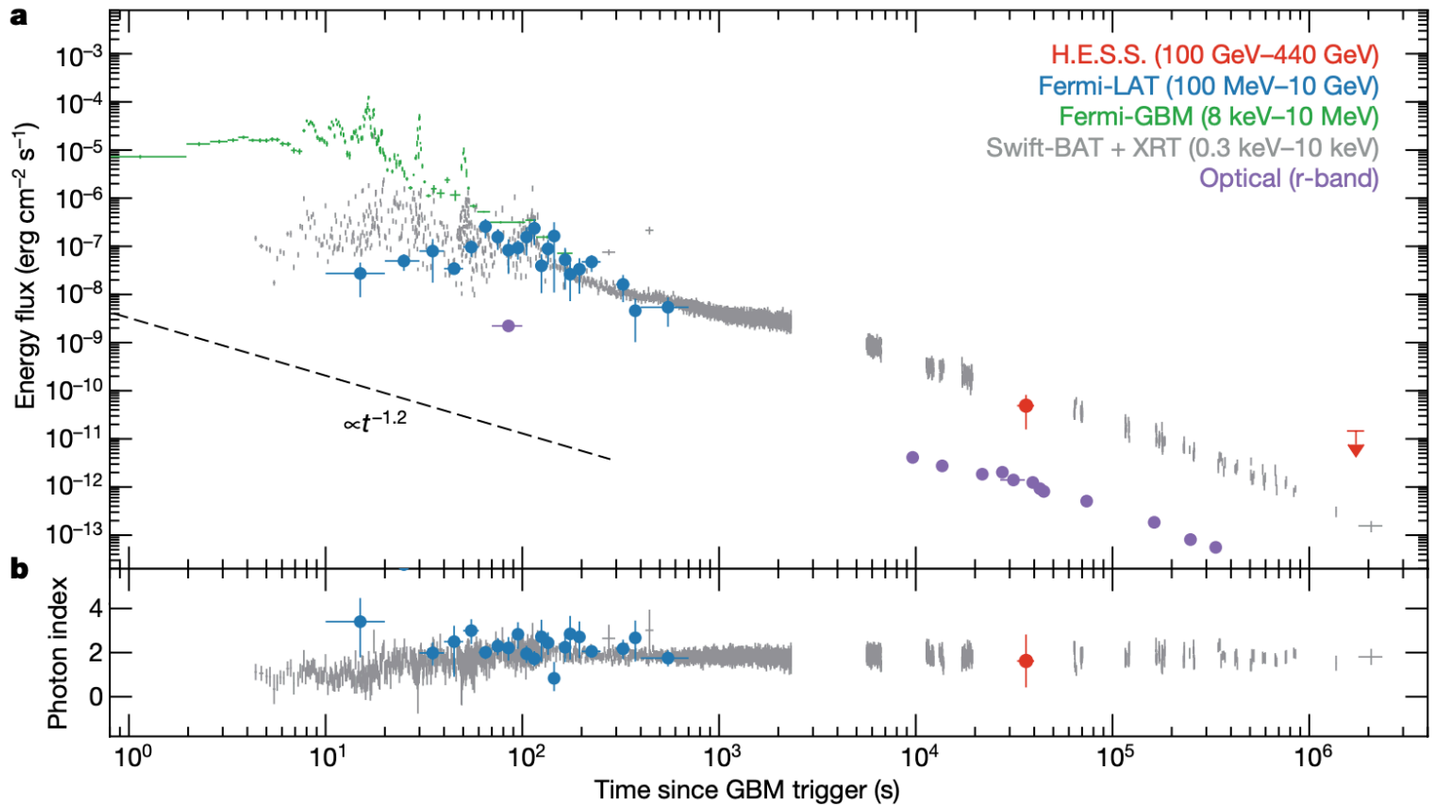
Gill & Granot, Galaxies 2022, 10, 74

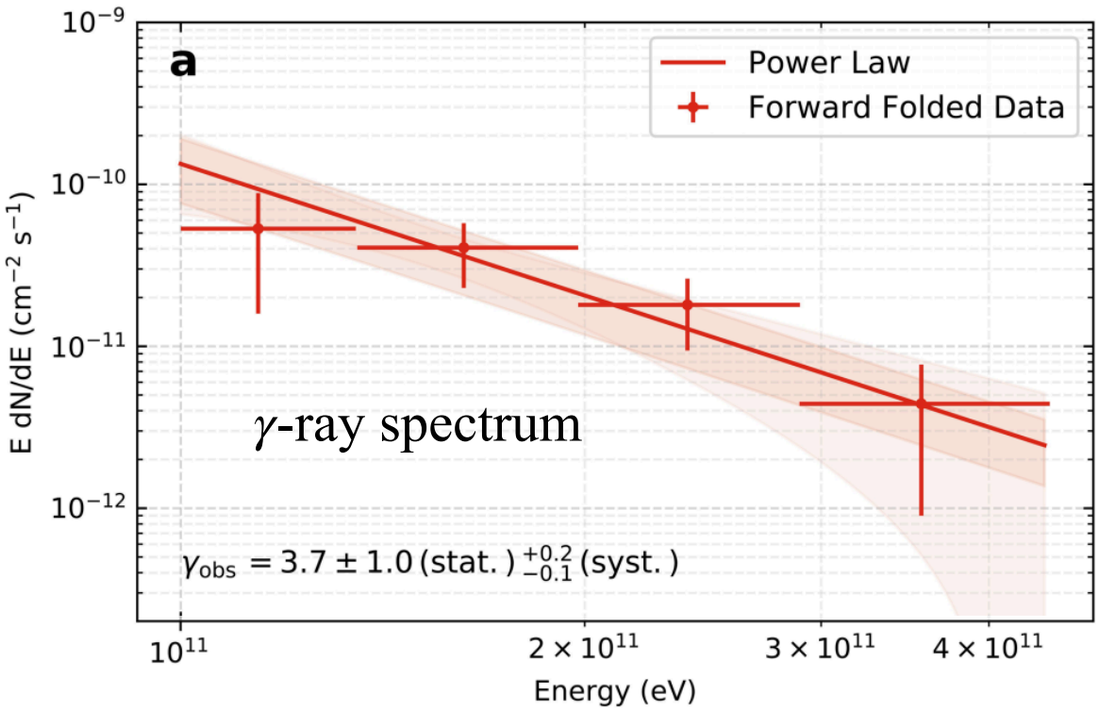
Extra-galactic objects

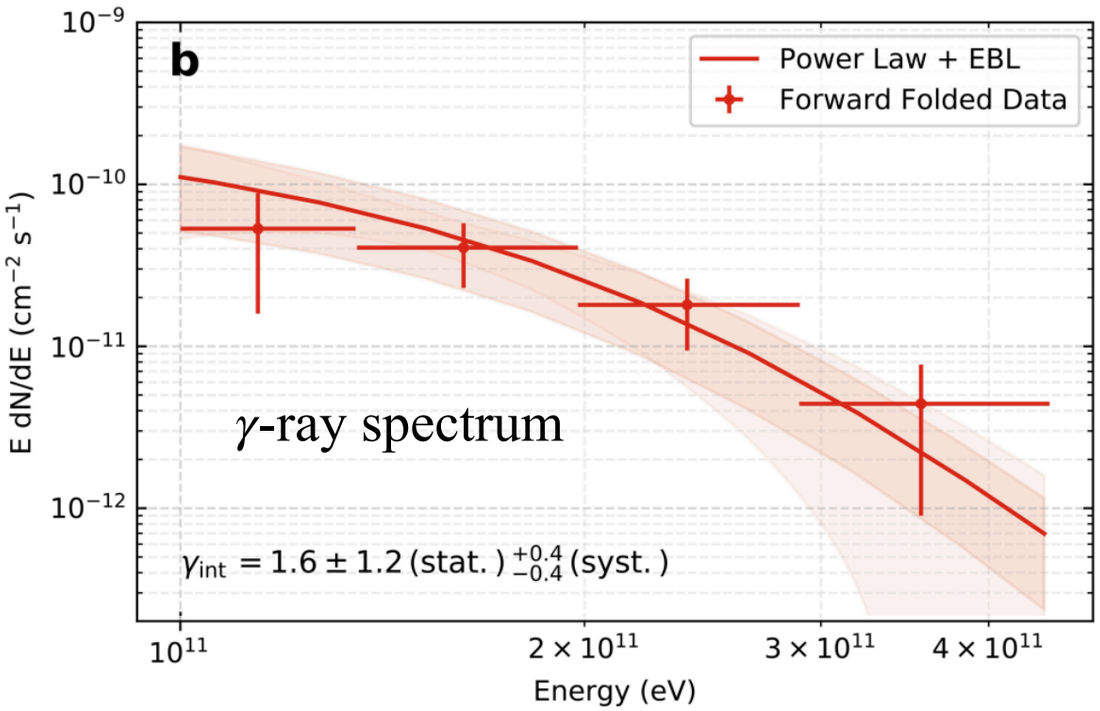
GRB 180720B



Light curve (Radio to TeV γ-rays)





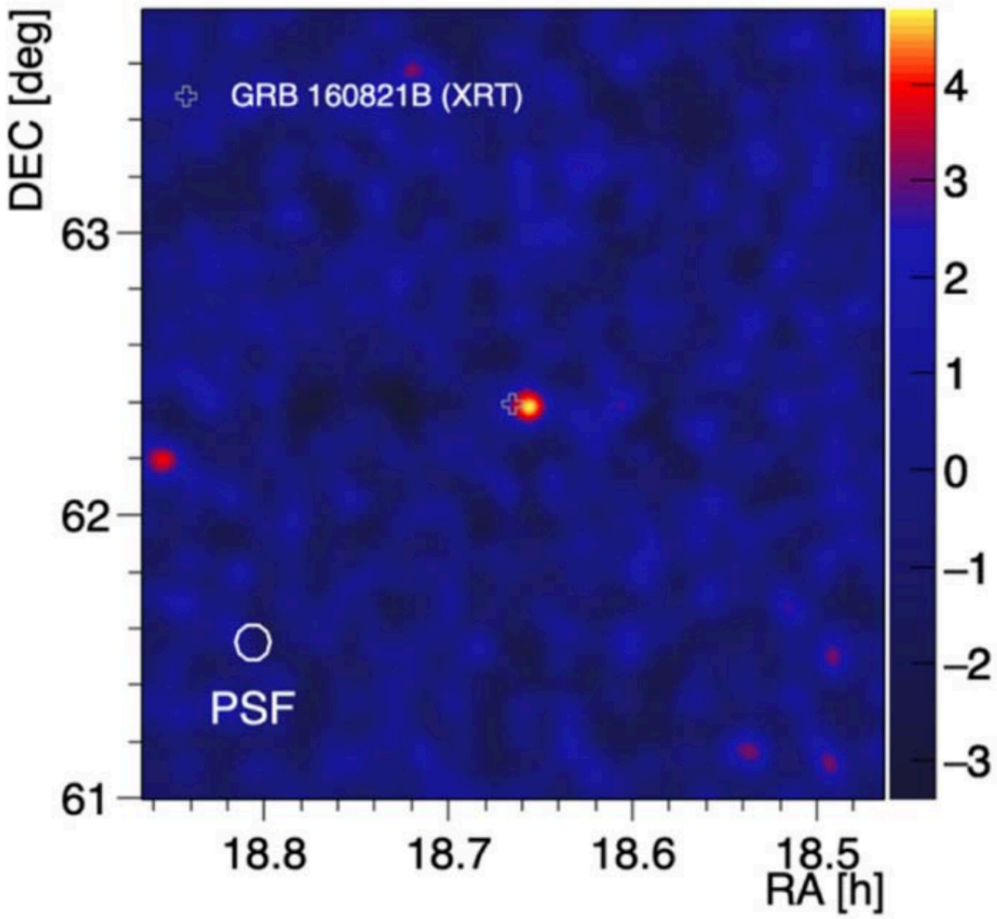


HESS Collaboration, 2019, Nature, 464, 21

Extra-galactic objects

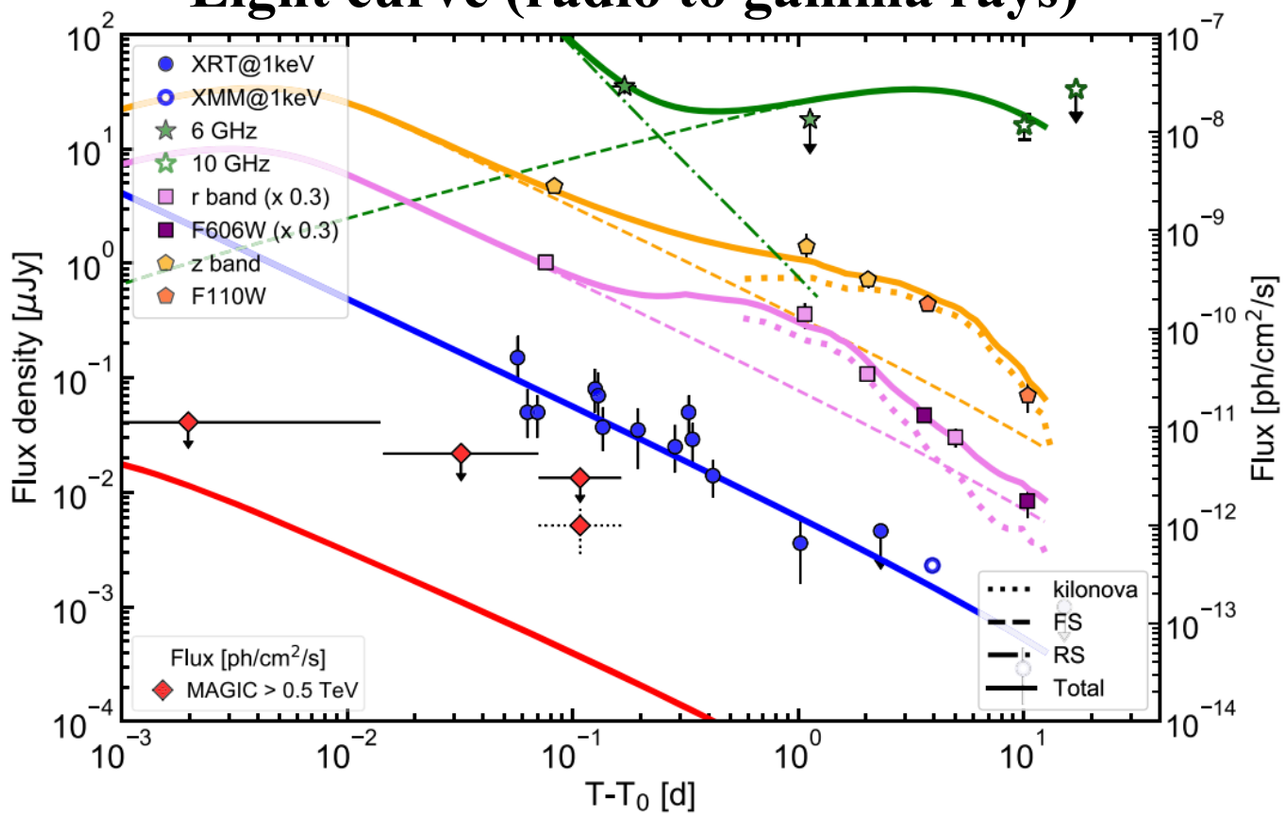
GRB 160821B

Significance map in γ-rays

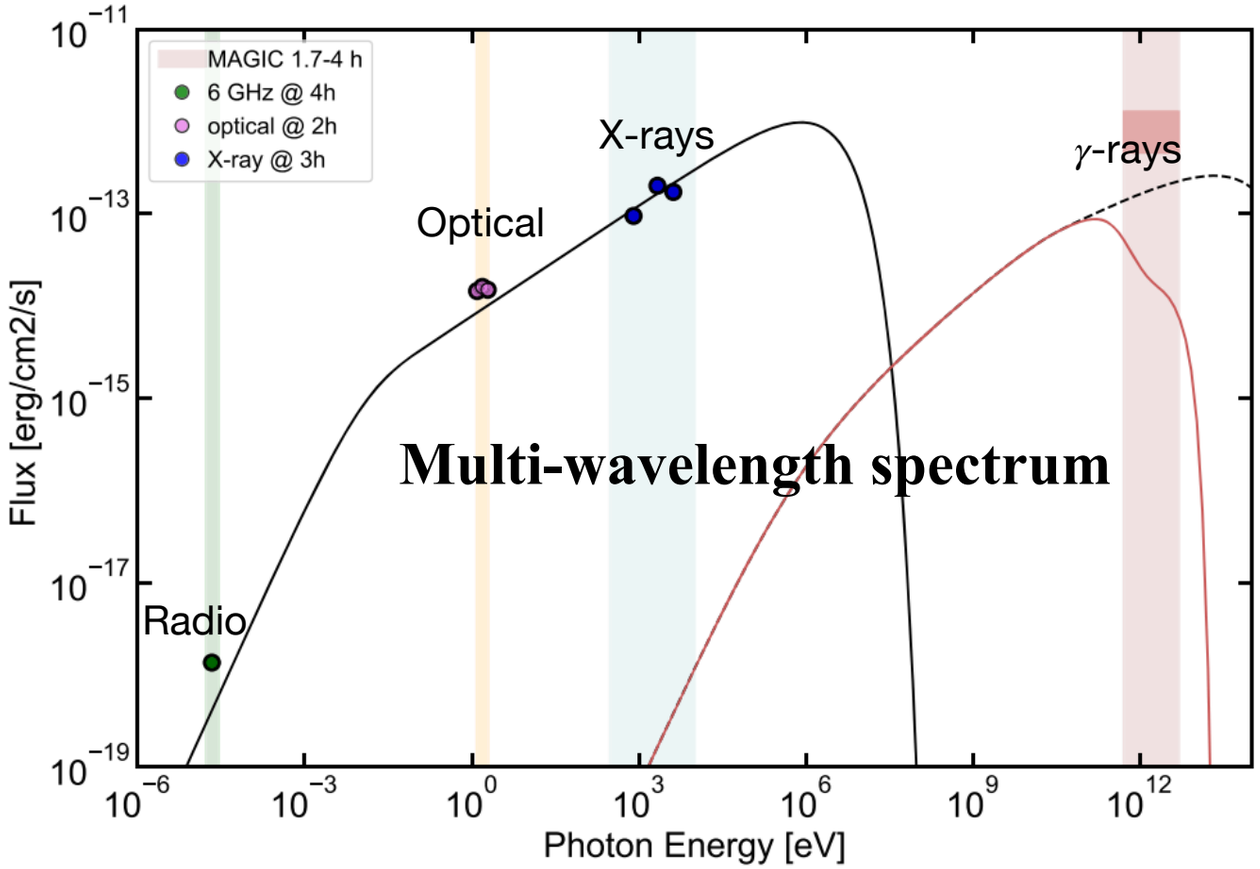


MAGIC Collaboration

Light curve (radio to gamma rays)



Multi-wavelength spectrum



MAGIC Collaboration, The Astrophysical Journal, 2021, 908, 90

Next, we will learn about the emission of gamma rays and neutrinos from astrophysical objects.